

EDUCATION AND EMPLOYMENT

- 1 SEP 2024 Postdoctoral Scholar, **Leiden University**, Leiden, the Netherlands
– (ONGOING) Supervisor: Jan Vonk
- 1 AUG 2021 CRM-ISM Postdoctoral Fellow, **McGill University**, Montreal, Canada
– 31 JUL 2024 Supervisors: Henri Darmon, Eyal Goren, and Michael Lipnowski
- JUN 2021 PhD in MATHEMATICS, **Stanford University**, Stanford, USA
Advisors: Akshay Venkatesh, Dan Boneh, and Ravi Vakil
Thesis: “Isogeny Graphs, Zero-cycles, and Modular Forms: Computations over Algebraic Curves and Surfaces”
- Nov 2016 MSc in MATHEMATICS, **University of Toronto**, Toronto, Canada
Advisor: Jacob Tsimerman
Thesis: “Field Extensions Generated by Kernels of Isogenies”
- APR 2015 Honours Bachelor of Science in MATHEMATICS, **University of Toronto**, Toronto, Canada

PAPERS AND PREPRINTS

1. “Hyperelliptic curves mapping to abelian varieties and applications to Beilinson’s Conjecture for zero-cycles” with Evangelia Gazaki. *Advances in Mathematics*, Vol. 487 (2026) 38 pgs ([DOI](#))
2. “Supersingular elliptic curves, quaternion algebras and applications to cryptography” with Eyal Z. Goren. *NATO Science for Peace and Security Series - D: Information and Communication Security*, Vol. 66: Abelian Surfaces and Isogeny-based Cryptography (2025) pp. 123–200 ([DOI](#))
3. “Rational configuration problems and a family of curves.” *Journal of Number Theory*, Vol. 269 (2025) pp. 370–396 ([DOI](#))
4. “On elements of prescribed norm in maximal orders of a quaternion algebra” with Eyal Z. Goren. *Canadian Journal of Mathematics*, Vol. 77, Issue 6 (2025), pp. 1938–1965 ([DOI](#))
5. “Torsion phenomena for zero-cycles on a product of curves over a number field” with Evangelia Gazaki. *Research in Number Theory*, Vol. 10, No. 35 (2024), 19 pgs ([DOI](#))
6. “Root Numbers of a Family of Elliptic Curves and Two Applications.” *Indagationes Mathematicae*, Vol. 35, Issue 3 (2024) pp. 555–569 ([DOI](#))
7. “Rational Equivalences on Products of Elliptic Curves in a Family.” *Journal de Théorie des Nombres de Bordeaux*, Vol. 32, No. 2 (2020) pp. 923–938 ([DOI](#))
8. “Supersingular Curves With Small Non-integer Endomorphisms” with Dan Boneh. In *Proceedings of the Fourteenth Algorithmic Number Theory Symposium*, ed. Steven D. Galbraith. *The Open Book Series*, Vol. 4, No. 1 (2020) pp. 7–22 ([DOI](#))
9. “Hypersurfaces passing through the Galois orbit of a point” with Shamil Asgarli and Chi Hoi Yip. 27 pgs. *Submitted* ([arXiv](#))

10. “On ℓ -torsion in degree ℓ superelliptic Jacobians over F_q ” with Wanlin Li and Eric Stubbley. 35 pgs. *Submitted* ([arXiv](#))
11. “Local and local-to-global principles for zero-cycles on geometrically Kummer K3 surfaces” with Evangelia Gazaki. 27 pgs. *Submitted* ([arXiv](#))
12. “An Arithmetic Variant of Raynaud’s Theorem” with Libby Taylor. 16 pgs. *Submitted* ([arXiv](#))

STUDENTS SUPERVISED

CURRENT	Pjotr Beerens: Master’s Thesis, Leiden
2025	Robbe Rietveld: Modular Arithmetic of Quaternion Norms , Bachelor’s Thesis, Leiden

RECENT PRESENTATIONS

<u>INVITED</u>	
DEC 2025	Leiden Mathematical Institute Master’s Colloquium
NOV 2025	Illinois State University Undergraduate Mathematics Colloquium
OCT 2025	Leiden University Mathematics Colloquium
JUL 2025	King’s College London Department of Mathematics
MAY 2025	The Canadian Discrete and Algorithmic Mathematics Conference (CanaDAM) minisymposium: Finite Fields and Finite Geometry
MAY 2025	Centre Interuniversitaire en Calcul Mathématique Algébrique (CICMA) Special Lecture
MAY 2025	Institut de Mathématiques de Jussieu–Paris Rive Gauche (IMJ–PRG) Number Theory Seminar
APR 2025	Leiden University AGNT Seminar
APR 2025	The Isogeny Club (online)
MAR 2025	University of Michigan Math Club
MAR 2025	University of Michigan GLNT Seminar
MAR 2025	University of North Carolina Chapel Hill Department of Mathematics
JAN 2025	Washington University in St. Louis AAG Seminar
JAN 2025	University of Pittsburgh ACG Seminar
DEC 2024	Groningen Algebra Seminar
NOV 2024	KULB-seminars (KU Leuven)
OCT 2024	Intercity Number Theory Seminar (Leiden University)
SEP 2024	Utrecht Algebraic Geometry Seminar
APR 2024	Front Range Number Theory Day (CU Boulder)
MAR 2024	McGill Arithmetic Geometry Graduate Student Seminar
DEC 2023	Canadian Mathematical Society (CMS) Session: Number theory by early career researchers
AUG 2023	Applied Mathematics, Modeling and Computational Science (AMMCS) Session: Computational Number Theory
FEB 2023	Fields Number Theory Seminar (online)
DEC 2022	Canadian Mathematical Society (CMS) Session: Diophantine Arithmetic Geometry and Number Theory
DEC 2022	Washington University in St. Louis AAG Seminar

APR 2022	American Mathematical Society (AMS) Session: Explicit Methods in Modularity (online)
MAR 2022	Montréal Online Biweekly Inter-University Seminar on Analytic Number Theory (MOBIUS ANT)
NOV 2021	Ottawa-Carleton Number Theory Seminar
SEP 2021	Québec-Vermont Number Theory Seminar
MAR 2021	University of Virginia Number Theory Seminar
FEB 2021	Québec-Vermont Number Theory Seminar
<u>CONTRIBUTED</u>	
AUG 2025	Banff International Research Station (BIRS) Workshop: Isogeny Graphs in Cryptography
APR 2025	Leiden Preprint Algebraic geometry and Number theory Cake (PANCake) Seminar
NOV 2024	Discrete, Interactive and Algorithmic Mathematics, Algebra and Number Theory (DIAMANT) Symposium
JUN 2024	Canadian Number Theory Association
OCT 2023	Maine-Québec Number Theory Conference
APR 2023	Postdocs at Centre de recherches mathématiques (CRM) Seminar
OCT 2022	Québec-Maine Number Theory Conference
JUL 2022	Park City Mathematics Institute
OCT 2021	Maine-Québec Number Theory Conference
JUL 2020	Fourteenth Algorithmic Number Theory Symposium (online)

TEACHING POSITIONS

JAN 2025	Instructor at LEIDEN UNIVERSITY
– PRESENT	Mathematics for Quantum Physics (QIST4100), Fall 2025 Topics in Algebraic Number Theory (4373TOANT), Winter 2025
JAN 2022	Instructor at MCGILL UNIVERSITY
– APR 2023	Number Theory / Honours Number Theory (MATH 346/377), Winter 2023 Discrete Structures (MATH 240), Winter 2022
JAN 2017	Teaching Assistant / Course Assistant at STANFORD UNIVERSITY
– MAR 2021	Linear Algebra and Multivariable Calculus (MATH 51), Fall 2018, Fall 2020 Differential Equations (MATH 53), Spring 2017 Modern Mathematics: Discrete Methods (MATH 62DM), Winter 2018, Winter 2020 Applied Matrix Theory (MATH 104), Spring 2020 Linear Algebra and Matrix Theory (MATH 113), Winter 2021 Groups and Rings (MATH 120), Fall 2017, Spring 2019
SUMMER 2018, 19	Mentor at CANADA/USA MATHCAMP Developed curriculum for and taught 25 hours of course material each summer.
SEP 2012	Teaching Assistant at UNIVERSITY OF TORONTO
– AUG 2016	Calculus (MAT135), Fall 2012 Calculus! (MAT137), 2013-14 Engineering Calculus (MAT187), Winter 2013 Groups, Rings, and Fields (MAT347), 2014-15, 2015-16.

OUTREACH, MENTORSHIP, AND SERVICE

Nov 2024	Institute Council (Instituutsraad) Member
– PRESENT	Mathematics Institute, Leiden University
2024	Mathematics Foundation of America (MFOA) Planning Committee
JUL 2024	Program Committee: Algorithmic Number Theory Symposium
JAN 2024	Organizer: JMM Special Session on Explicit Computations with Stacks
SEP 2023	Research Supervisor at MCGILL UNIVERSITY
– APR 2024	Supervising two undergraduate students studying elliptic curves, and one studying quaternion algebras (MATH 470)
SUMMER	Academic Coordinator at CANADA/USA MATHCAMP
2021	A 5-week long summer program for mathematically advanced high school students. Designed the academic schedule for the program, invited guest speakers, and coordinated teacher training and development.
MAR – MAY	Teaching Assistant at ARIZONA WINTER SEMESTER
2021	Developed problem sets and held office hours for a 6-week course on p -adic numbers and p -adic modular forms.
MAR 2018	Teaching Assistant at GEOMETRY OF REDISTRICTING (San Francisco) 2-day conference hosted by the Metric Geometry and Gerrymandering Group. Facilitated group activities designed to train high school math teachers and college professors in the mathematics of gerrymandering, voting, and apportionment.
APR 2017	Directed Reading Program Mentor at STANFORD UNIVERSITY
– JUN 2021	Worked with eight undergraduate students on an individual basis; one meeting per week for 10 weeks each
MAR 2014,	Lecturer at MARCH BREAK MATH ACADEMY, Toronto, Canada
MAR 2015	5 days, 7 hours per day. Hosted by University of Toronto Schools.
VARIOUS	Guest Speaker Gunn High School Math Circle, Palo Alto, Feb 2020 The Wilberforce School, Princeton, Mar 2019 University of Toronto Mississauga Math Circle, Apr 2015, Nov 2015, Feb 2016 Pre-Concert Lecture for Aradia Ensemble, Toronto, Oct 2014

GRANTS AND AWARDS

2021-23	CRM-ISM Postdoctoral Fellowship
JUN 2021	Centennial Teaching Assistant Award, Stanford University
JUL 2020	Selfridge Prize, Algorithmic Number Theory Symposium

COMPUTER SKILLS

WORKING Sage, Mathematica, Magma, Python

BASIC HTML, CSS, PHP, Javascript, PARI/GP

My Github page (<https://github.com/jonathanrlove>) contains a sample of my computational work, including the following projects:

- Computing supersingular isogeny graphs and identifying curves that have non-integer endomorphisms of small degree (supporting the paper “Supersingular Curves With Small Non-integer Endomorphisms”)
- Producing rational equivalences on certain products of elliptic curves (supporting the paper “Torsion phenomena for zero-cycles on a product of curves over a number field”)
- Computing cusp forms over function fields, using an algorithm that tests for isomorphisms between rank 2 vector bundles (supporting my PhD thesis)